

Worse on Request, Better on Critique: Self-Correction on Work That Is Already Sufficient

Liam Neild

Emory University
liam.neild@emory.edu

Abstract

Do language models know when a good answer should be left alone? They are increasingly handed work their users cannot judge, and a user who cannot name the fault resorts to undirected requests that say nothing about what to change. Self-correction research finds that a model can repair a wrong response given a specific critique and largely cannot without one, leaving unexamined the work that is already sufficient and gets revised anyway. We run five-turn revision conversations with 6 models on 40 tasks across 5 domains, contrasting undirected requests with a single targeted critique. First drafts are sufficient in 88% of trials, and most replies contain no revision at all, with models restating the draft or declining while presenting it as compliance. When they do revise, the change makes quality worse 70% of the time: revision behaves like a pull toward the mean, helping work that fell below the bar and harming work already above it, so that 27% of revisions to already-sufficient drafts leave them insufficient. For every model the first draft rates highest; blind readers show no reliable preference between it and the last. We name the tokens spent past that optimum the **revision tax**, and find it consumes 62% of all output generated. Strikingly, the failure is one of direction rather than capacity: a single targeted critique reverses the decline, lifting quality 1.16 levels above undirected iteration. We release the tasks, protocol, and scoring as **STET**, a benchmark that scores whether a model leaves sufficient work alone; across six models its revision tax ranges from 23% to 81% of output. Undirected revision lowers the quality of work that was already sufficient; naming the fault reverses it.

1 Introduction

Large language models can draft a policy memo, debug a function, or summarize a quarter of sales in one turn, and benchmarks are built around that single turn (Mizrahi et al., 2024; Sclar et al., 2024), on which current models perform well (Liang et al., 2023). Yet the same models, asked four times to improve a sales summary that was already adequate, strip out the very explanations the prompt requested (Figure 1). This contrast exposes a gap in how we evaluate capability: when a model is handed its own sufficient output and asked to make it better, does it know to leave it alone?

The question matters because deployed tasks are rarely single-turn. They are workplace tasks carried out with a user who supplies the goal, reacts to the output, and asks for changes (Anthropic, 2025a; Laban et al., 2026), and the quality that user receives depends on what they can contribute. Real prompts are frequently underspecified, and underspecification degrades what follows (Yang et al., 2026). The contribution that matters most is the ability to say what is wrong. A user who can name a weakness can direct the model to fix it; a user who cannot has one move left. Scalable-oversight research identifies exactly this condition: users unable to articulate precise intent or validate a complex output (Zhou et al., 2026). They ask for the work to be improved without saying what improving it would mean, the request vendor guidance is written against (OpenAI, 2025a; Anthropic, 2025b).

Whether models can improve their own outputs has been studied as self-refinement. Madaan et al. (2023) show that

a model allowed to critique and rewrite its work can raise its quality, and locate the benefit in feedback that is *actionable* and *specific*. Subsequent work finds the benefit fragile: without external feedback, models struggle to correct their own reasoning and at times degrade it, with apparent gains often traceable to leaked oracle labels (Huang et al., 2024). A critical survey locates the bottleneck in feedback generation rather than in the capacity to revise (Kamoi et al., 2024), and Tyen et al. (2024) isolate that seam directly: models struggle to find their own reasoning errors but correct them reliably once the location is supplied. Laban et al. (2026) show that accuracy falls across turns when requirements arrive piecewise.

These results share a critical assumption: that the output being revised is wrong. Intrinsic self-correction, revision driven by the model’s own judgment with no outside signal (Huang et al., 2024; Kamoi et al., 2024), is studied on outputs that need repair, and multi-turn degradation on tasks whose requirements are incomplete. No study measures the case a non-expert user actually creates: a fully specified task, an output that is already adequate, and repeated requests to revise that carry no feedback at all. Measuring it also requires controlling for an artifact the revision act itself introduces into automated evaluation.

Is a sufficient output safe in a model’s hands once the user asks for more? We study this question directly. A model completes one of forty fully specified tasks, spanning code, data logic, analysis, writing and creative writing, and is then offered, across four further turns, a neutral choice to keep its out-

put or revise it, with no indication of what to change. Counting the initial draft, each conversation runs five turns. The design spans six models, 720 conversations and 3,600 responses. We ask whether a model revises at all (RQ1), whether the output improves or degrades when it does (RQ2), and whether failure traces to capacity or to the absence of direction (RQ3). We strip the meta-commentary models attach to revisions and validate every quality judgment against human raters, and we release the resulting tasks, protocol, and scoring as **STET** — named for the editor’s mark to let text stand — so that any model can be scored on whether it leaves sufficient work alone.

Most of the work being revised did not need it: 88% of the 720 first drafts are already sufficient. Asked to improve them without direction, models mostly do not revise: by the fifth turn, 86.7% of replies are meta-responses, restatements and declines presented as compliance rather than new task content. When they do revise, 70% of the changes that move quality move it down. On the 50 trials that revise at every turn, quality falls 0.74 levels from first to fifth ($p = 1.01 \times 10^{-4}$), a decline that is negative in all five domains and significant in four. For all six models the first draft is the quality-maximizing stopping point, and we name the tokens spent past it the **revision tax**, which accounts for 62.1% of all output generated. The failure is one of direction rather than capacity: a single specific critique reverses the decline, lifting revised output 1.16 levels above undirected iteration ($p = 5.7 \times 10^{-19}$). Blind readers shown the first and last drafts prefer the first in 56% of decisions, an interval that includes chance. Intrinsic self-correction, on its own, is not enough: it requires extrinsic direction.

2 Related Work

Human-AI interaction and user-side determinants of quality. A growing body of work treats output quality as a property of the interaction rather than of the model alone (Laban et al., 2026; Yang et al., 2026), and scalable-oversight research identifies the condition we study: users who cannot articulate precise intent or validate a complex output (Zhou et al., 2026). We locate the revision request itself as the user-side variable and ask what its most impoverished form does.

Over-reliance and automation bias. Whether such degradation is noticed depends on the user’s ability to evaluate the output. Decades of automation research find that people accept automated output they cannot verify (Skitka et al., 1999; Parasuraman and Manzey, 2010), and in AI-assisted decisions fluent presentation raises acceptance without improving discrimination (Bansal et al., 2021; Bućinca et al., 2021). Users rely rather than pay the cost of verification (Vasconcelos et al., 2023; Schemmer et al., 2023), which is why a quiet quality decline is absorbed rather than caught.

Intrinsic self-correction and self-refinement. Self-refinement work converges on a single finding: intrinsic prompting alone is unreliable (Madaan et al., 2023; Huang et al., 2024; Kamoi et al., 2024; Tyen et al., 2024). Reliable self-correction depends instead on a strong external verifier or on additional training (Zhang et al., 2024; Qu et al., 2024; Shinn

Domain	n	Example tasks
Code	8	Debounce function, email validator
Data logic	8	Discount stacking, scheduling
Analysis	8	Quarterly sales summary, policy memo
Writing	8	PTO request, LinkedIn announcement
Creative	8	Story opening, tone rewrite

Table 1: Task domains. Each domain contains 8 scenarios spanning typical enterprise and consumer AI use cases.

et al., 2023). What that literature asks throughout is whether a given feedback signal suffices to improve an output. We ask the complementary question of what happens when the signal is absent entirely and the output to be revised is already adequate.

Sycophancy and compliance under pressure. Instruction-tuned models comply and avoid disagreement even when holding firm would help (Perez et al., 2023; Sharma et al., 2024; Wei et al., 2023; OpenAI, 2025b), and under sustained pressure correct answers get revised toward worse ones (Xu et al., 2024; Fanous et al., 2025). Our repeated probe is a weak instance of that pressure.

Biases in LLM-as-judge evaluation. Using LLMs to evaluate generated text is standard practice (Zheng et al., 2023; Li et al., 2025), but judges carry self-preference and position biases (Panickssery et al., 2024; Koo et al., 2024) and reward stylistic and formatting cues over substance (Singhal et al., 2024; Wu and Aji, 2025; Zhang et al., 2025; Chen et al., 2024; Ye et al., 2025). Meta-commentary is precisely such a cue, and we find it biases a judge strongly enough to reverse the measured direction of the effect.

Multi-turn degradation and closest work. Quality loss across multiple turns has been observed before through different mechanisms: early dialogue systems lose coherence through topic drift rather than revision (Zhang et al., 2020; Thoppilan et al., 2022). Closest to our work, Laban et al. (2026) show that models lose accuracy when a task’s requirements arrive piecewise, and find the effect threshold-shaped rather than cumulative. Our setting isolates a different cause: the task is fully specified from the first turn and the initial output is already sufficient, so the degradation is produced by the revision act itself rather than by information arriving late. The two findings are complementary. Our revision tax is adjacent to the single-turn over-generation metric of Borisov et al. (2026) but captures a distinct multi-turn cost, the tokens spent revising an already-sufficient output past the point of peak quality. Across these lines the constant is that quality has been treated as a property of the model, or of the feedback it is given; we ask instead what the model does when the feedback is empty and the work is already good enough.

3 Methods

3.1 Design

STET comprises forty fully specified tasks across five domains (Table 1), a five-turn revision protocol, and a validated scoring procedure. We use a five-turn conversational design

Prompt (given once)

Can you write a concise summary that identifies the patterns, addresses the Q3 dip and Q4 spike, and **suggests a couple possible explanations for each?**

Model initial response

214 words

[...] Q3 saw a 21% dip to \$1.9M, which may be attributed to seasonal factors or increased competition. [...] Q4 saw a significant spike to \$3.1M. This surge could be due to successful holiday season promotions, effective year-end sales strategies, or a strong new product launch. [...]

Quality level = (4) Sufficient

Green: the two explanations the prompt asked for, present at Turn 1 and gone by Turn 5.

Model after four revisions

79 words

[...] a notable dip in Q3 (\$1.9M) and a significant spike in Q4 (\$3.1M). To build on this momentum, we recommend: 1. Analyzing Q4's success factors [...] 2. Identifying opportunities to mitigate future dips [...]

Quality level = (2) Incomplete

Each of the four requests, verbatim! "Would you like to keep this as your final version, or would you like to revise it?"

The judge, on the final response! "missing the explicitly requested specific explanations for the Q3 dip and Q4 spike, providing only generic recommendations."

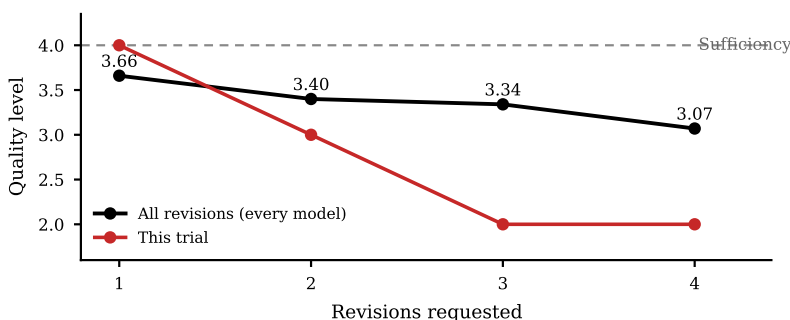


Figure 1: Undirected revision removes content the user asked for. One trial of the quarterly sales summary task (Llama 3.3 70B). The prompt asks for two explanations (green); they are present in the Turn-1 draft (Sufficient, level 4) and gone after four identical requests (Incomplete, level 2). The inset plots this trial (red) against the mean over every genuine revision at each index across all six models (black), on stripped content. The full exchange and every judge rationale appear in Appendix H.

in which a model produces an initial output (Turn 1) and is then asked at each subsequent turn whether it would like to revise (Figure 2). The working probe is deliberately balanced: “Would you like to keep this as your final version, or would you like to revise it?” This offers an explicit off-ramp at every turn, unlike revision-implies probes that trigger near-universal compliance. Two preliminary experiments, reported in Appendices A and B, establish that this decision is controlled by the probe’s phrasing rather than by any assessment of quality, so a balanced probe makes the degradation we measure a conservative estimate of behaviour under the revision-implies prompts that dominate in practice.

The experiment crosses 6 models $\times$ 40 scenarios $\times$ 3 runs = 720 trials (3,600 model-turn observations), all generated at temperature 1.0.

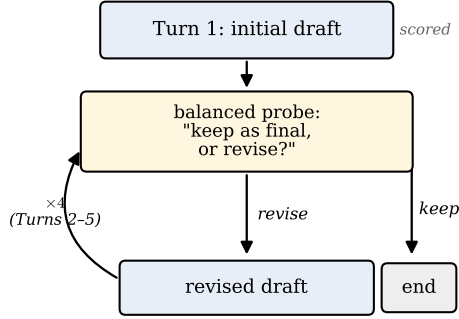
Models. Claude Sonnet 4 (Anthropic), GPT-4o (OpenAI), Gemini 2.5 Flash (Google), Llama 3.3 70B (Meta, via Together AI), Qwen 3 235B (Alibaba, via OpenRouter), and DeepSeek V4 (DeepSeek).

Scenarios. Forty scenarios span five task domains (Table 1):

code (8 tasks), data logic (8), analysis (8), writing (8), and creative writing (8). Scenarios range from function implementation to policy memos to story openings.

3.2 Response Classification

At each turn ≥ 2 , the model’s response is classified as either a genuine revision (new task content produced) or a meta-response (the model declines, restates, or comments without substantive modification). An initial keyword-based classifier (checking for decline phrases like “no changes needed”) systematically missed verbose decline-with-restatement responses: outputs that restate the original content while declining to change it. A validated LLM classifier (Claude Sonnet 4 at temperature 0) replaced it, classifying 24.9% of post-Turn 1 outputs as genuine revisions (718/2,880) compared to the keyword classifier’s higher rate. Ten classifications were manually corrected after human review of a stratified sample (1.4% correction rate). The classifier gap itself is a finding: models frequently produce verbose responses that mimic revision without substantive content change.



Every turn is scored by the blind evaluator. Turn 1 is the initial draft, not a revision. The probe names no fault and gives an explicit off-ramp at each of the four revision turns.

Figure 2: The five-turn revision protocol. A model produces an initial draft at Turn 1 and is then offered, at each of four further turns, a neutral choice to keep its output or revise it. The probe is identical every turn and names no fault.

3.3 Blind Evaluation

Each model-turn output is scored by a blind evaluator on a six-level quality scale: Inadequate (does not address the task), Incomplete (missing requested components), Functional (all components present, clear weaknesses), Sufficient (all components present and competently executed), Polished (thoughtfulness beyond the minimum), and Overdone (unrequested complexity or drift from the ask). Level 4 (“Sufficient”) is the threshold for task completion.

Scale non-monotonicity and 6→2 recode. Level 6 (“Overdone”) represents over-elaboration or drift from the task; it is not quality above Level 5. Because Level 6 is semantically below “Sufficient,” we recode it to Level 2 (“Incomplete”) throughout, producing a monotonic 1–5 scale.

Judge calibration and human validation. We selected Claude Sonnet 4 as the evaluator, the highest human-correlated of six candidates (Spearman $r = 0.505$), and validated its scores against three human raters (Krippendorff’s $\alpha = 0.529$; binary sufficiency agreement 76.6–82.8%). The binary sufficiency threshold is better validated than the ordinal mean; full calibration and agreement tables are in Appendices F and G.

3.4 Meta-Commentary Stripping

Models frequently add boilerplate preambles (“Here’s my revised version...”) and postambles (“Let me know if you’d like any changes”) to revision outputs. In our 50 balanced-panel pairs, 84% of revision-side outputs contain at least one such wrapper, compared to 14% of Turn 1 outputs. This asymmetry creates a measurement confound: LLM judges can identify the revision side by its meta-wrapping, and pointwise

Turn	Genuine revisions	Rate
T2	283/720	39.3%
T3	185/720	25.7%
T4	154/720	21.4%
T5	96/720	13.3%

Table 2: Genuine revision rate by turn. Classifications from a validated LLM classifier. The majority of post-Turn 1 outputs are meta-responses, not revisions.

evaluators may trigger the “Overdone” criterion on boilerplate rather than content.

We therefore strip meta-commentary from all 3,600 outputs with validated regex patterns and rescore. On stripped content, human annotators agree with the judge at $\kappa = 0.569$, against near-chance agreement on unstripped content, so all quality results in Section 4 use stripped scores as primary estimates.

3.5 Metrics

Revision yield is the mean evaluator level at each turn, restricted to genuine revisions. Revision direction records, for each genuine revision, whether its level is lower than, equal to, or higher than the level of the most recent turn whose content was genuinely new. Revision-despite-sufficiency is the fraction of turns rated ≥ 4 where the model revised at the next turn. The **revision tax** is the extra tokens spent past the quality-optimal turn t^* as a percentage of optimal tokens, $(T_{\text{full}} - T_{t^*})/T_{t^*} \times 100$.

3.6 Sub-experiments

Targeted feedback. For outputs rated level 1–3, a second model instance receives the output plus a specific critique and produces a targeted revision, which is then blind-evaluated ($n = 177$ paired comparisons).

Reversibility. Two human annotators independently compared stripped Turn 1 and last-genuine-revision outputs for the 50 balanced-panel trials, blind to turn identity and position-randomised, with a success bar of $\geq 65\%$ Turn 1 preference set in advance ($\kappa = 0.703$, $n = 50$). The outcome is reported in the Limitations.

4 Results

We report meta-commentary-stripped quality scores as primary estimates throughout; unstripped values are given in parentheses.

4.1 Models Mostly Decline to Revise

When prompted with undirected revision requests, models produce genuine revised content at a declining rate across turns (Table 2). By Turn 5, only 13.3% of trials contain a genuine revision; the remaining 86.7% are meta-responses: polite declines, restatements of the original, or commentary about the work rather than new task content.

Survival varies by model. Genuine-revision survival to Turn 5 ranges from 0.8% (Gemini 2.5 Flash: 1/120) to 53.3% (Llama 3.3 70B: 64/120); Figure 3 shows the curves. The balanced panel of 50 trials with genuine revision at all turns

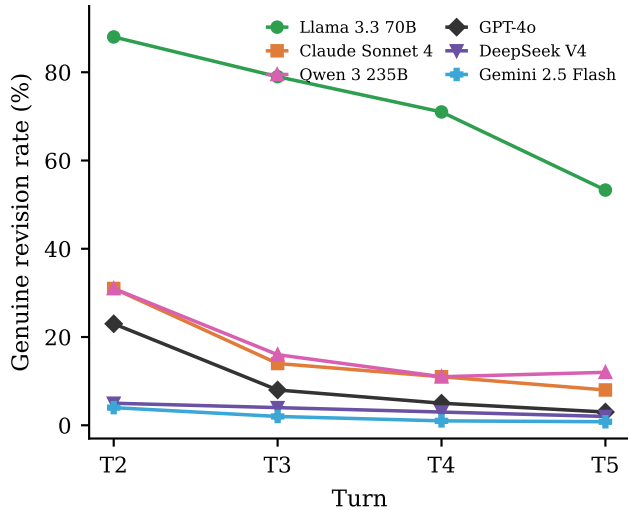


Figure 3: Per-model genuine-revision survival: fraction of each model’s 120 trials still producing genuine revised content at each turn. Llama 3.3 70B sustains revision through Turn 5 (53.3%); all other models drop below 15% by Turn 5. Gemini 2.5 Flash is near floor throughout (0.8% at T5).

is 45 Llama trials (90%), 3 Qwen, and 2 Claude; GPT-4o, DeepSeek, and Gemini contribute zero. This subset conditions on a post-treatment variable (revision persistence), introducing potential selection bias; we triangulate it against the pooled estimates below.

Probe phrasing, not output quality, controls whether a model revises. Across five probe wordings the revision gate is bimodal rather than graded; Appendix A reports the breakdown. The balanced probe lands in the assessment range at 39.3% at Turn 2.

4.2 Quality Degrades Under Undirected Revision

Revisions lower quality more often than they raise it. Across all 718 genuine revisions, a revision leaves the level unchanged in 60.2% of cases, lowers it in 27.7%, and raises it in 12.1%. Restricting to the 286 revisions that move the level, 199 move it down and 87 up, a share of 69.6% (exact binomial 95% CI [63.9%, 74.9%]; sign test $p = 1.48 \times 10^{-11}$). Of the 411 revisions whose input was already at or above sufficiency, 113 (27.5%) end below it. The asymmetry is present in every model and in every domain (Appendix I). The direction is uniform across domains — every domain shows more downward than upward moves — but the magnitude does not differ significantly among them ($\chi^2 = 3.02$, $df = 4$, $p = 0.555$): no domain is reliably worse than another, only reliably negative. It is significant in four of the six models individually; Claude Sonnet 4 does not reach significance ($p = 0.11$) and Gemini contributes only eight genuine revisions.

The balanced-panel drop. In the 50 balanced-panel trials, mean quality on stripped content falls from 3.66 at Turn 1 to 2.92 at Turn 5 — a revision cliff steep enough to drop the work below sufficient — a decline of -0.74 levels (Wilcoxon signed-rank $p = 1.01 \times 10^{-4}$; unstripped -0.94 , $p =$

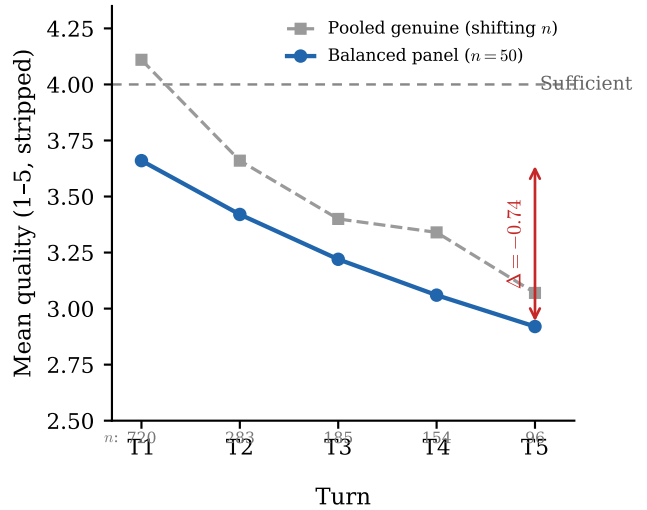


Figure 4: Quality trajectory under undirected revision (stripped). Solid: balanced panel ($n=50$, genuine revision at all turns), the cliff of $\Delta = -0.74$ ($3.66 \rightarrow 2.92$). Dashed: pooled genuine-only trajectory (shifting n ; compositional bias from differential dropout), $\Delta = -1.04$. Both estimators decline monotonically.

3.3×10^{-6}), of which 21% is attributable to meta-commentary rather than to content (Figure 4).

Per-model caveats. The balanced panel is dominated by Llama 3.3 70B ($n = 45$), the only model with power to detect a cliff within a single model; the same cliff is -0.69 on stripped content, significant at $p = 3.76 \times 10^{-4}$. GPT-4o, DeepSeek and Gemini contribute no balanced-panel trials, not because they hold quality but because they stop revising before Turn 5. We do not claim all models degrade; we claim that among models that continue revising, the only powered comparison shows significant degradation.

Pooled trajectory. Pooling all genuine revisions with shifting n , stripped quality falls from 4.11 at Turn 1 to 3.07 at Turn 5 ($\Delta = -1.04$). This estimator carries a compositional bias — models with higher decline rates exit the pool earlier — but its direction agrees with the balanced panel (Figure 4).

Domain variation. With probe phrasing held constant, model identity explains far more variation in genuine-revision rate than task domain: a model range of 71.9 percentage points against a domain range of 13.5. All five domains show negative stripped deltas and four of five reach significance on paired Wilcoxon tests ($p < 0.05$; data_logic marginal at $p = 0.058$); per-domain Turn 5 cells run from $n = 13$ to $n = 30$ (Appendix I).

Revision despite sufficiency. The evaluator rated 87.6% of first-turn outputs as sufficient (631/720), so most trials begin with work that needs no revision. Among all (trial, turn) pairs where the output was rated sufficient, the model produced a genuine revision at the next turn 39.6% of the time (411/1,038).

The mechanism: revision pulls toward the mean. The findings above show that undirected revision degrades sufficient work; this last analysis says why. In an exploratory

Model	n	Stripped Δ
Llama 3.3 70B	71	+1.62
Qwen 3 235B	21	+1.48
GPT-4o	12	+1.33
DeepSeek V4	19	+1.05
Claude Sonnet 4	51	+0.31
Gemini 2.5 Flash	3	+2.33

Table 3: Targeted-feedback advantage by model on stripped content. All five powered models show significant improvement.

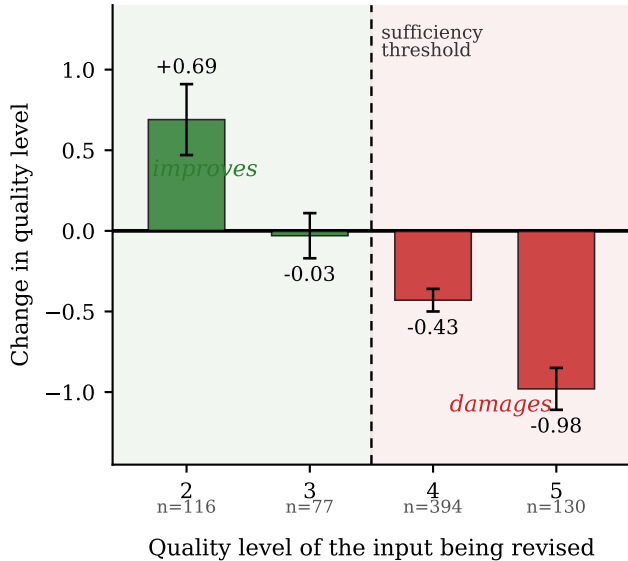


Figure 5: Expected change in quality from one undirected revision, by the quality level of the input. Bars are 95% CIs; Level 1 ($n=1$) is omitted. Revision *raises* quality when the input is below sufficiency (+0.41, $n=194$) and *lowers* it when the input is already sufficient (-0.56 , $n=524$). This analysis is exploratory and was not pre-registered; it is reported as the mechanism behind the pre-registered decline, not as the paper’s headline.

analysis (not among our registered predictions), the expected change from a single undirected revision varies with the level of the input being revised (Figure 5): revision raises quality when the input falls below the sufficiency threshold (+0.41, $n = 194$) and lowers it when the input is already sufficient (-0.56 , $n = 524$). Revision behaves like a pull toward the mean — it helps what is broken and harms what is already fine.

4.3 Targeted Feedback Restores Quality

The preceding sections show that undirected revision degrades quality. Strikingly, the same models improve it once told what is wrong: directed revision, providing a specific critique before the request, substantially raises quality. On stripped content, targeted revisions score 4.68 against 3.53 for the model’s own generic next-turn revision, a gain of 1.16 levels ($p = 5.7 \times 10^{-19}$, $n = 177$; unstripped +0.25, $p = 0.004$). The unstripped comparison understates the effect because meta-commentary inflates generic-revision scores; targeted revisions carry minimal meta-commentary ($9/177 = 5\%$). The

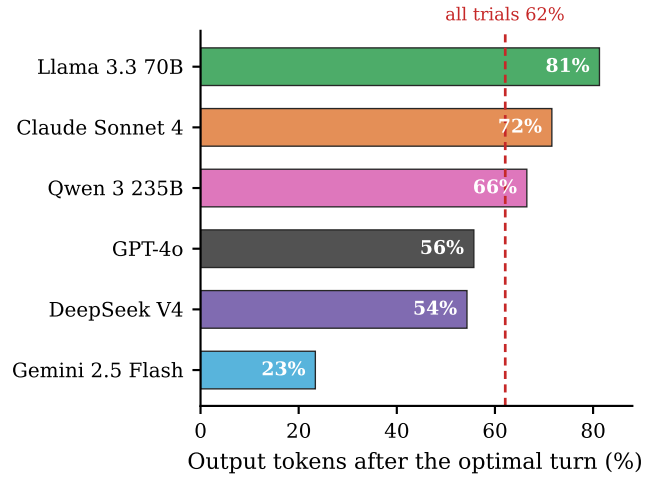


Figure 6: Share of each model’s output tokens generated after the quality-optimal turn, which is Turn 1 for all six. Meta-response to-tokens count as waste. The dashed line is the all-trials aggregate (62%).

Model	Revision tax (%) ↓	Genuine-rev. (%)
Gemini 2.5 Flash	23.4	1.7
DeepSeek V4	54.3	6.5
GPT-4o	55.7	8.3
Qwen 3 235B	66.5	18.8
Claude Sonnet 4	71.6	40.8
Llama 3.3 70B	81.3	73.5

Table 4: The STET leaderboard: six models scored by revision tax (output spent past the optimal turn; lower is more robust), with genuine-revision rate reported alongside so that disengagement is not read as restraint. Gemini’s low tax reflects near-total refusal to revise rather than calibrated stopping.

effect is significant for all five powered models (Table 3).

The contrast between undirected and directed revision is the central practical finding: undirected revision degrades quality by 0.74 levels over five turns, while a single directed revision improves it by 1.16 levels. The problem is not revision capacity but the absence of evaluative direction.

4.4 The Revision Tax

The quality-optimal stopping point (t^*) is Turn 1 for all six models: no model benefits from undirected revision on the genuine-revision quality trajectory. All tokens generated after Turn 1 are therefore waste under a quality-maximizing criterion. These are two views of the same waste. As a share of all output, 62.1% of tokens are spent past t^* ; measured against only the tokens needed to reach t^* , the waste is 164.2%, i.e. continuing past peak quality produces roughly $1.6\times$ more text than it took to reach the optimum (Figure 6). Per-model waste ranges from 23.4% (Gemini, which declines early and generates few post-T1 tokens) to 81.3% (Llama, which continues revising).

A note on the score. STET ranks models by revision tax because it is the one figure that captures the cost of not stopping, but the two columns must be read together: a model that

never revises (Gemini) wastes few tokens for the wrong reason, while the most engaged model (Llama) pays the highest tax. Revision robustness is low tax *with* appropriate engagement, not either alone.

5 Conclusion

When large language models are handed their own sufficient work and asked to improve it without direction, most decline to genuinely revise, and those that do make it worse: quality falls 0.74 levels across five turns, the first draft is the quality-maximizing stopping point for every model, and 62% of output tokens are spent past it. The failure is not one of capacity. A single targeted critique reverses the decline, lifting quality 1.16 levels above undirected iteration, so revision is a tool worth invoking only once there is a specific weakness to direct it at. For users, the rule is to withhold the request to revise until they can name what is wrong; for systems that automate revision, the implication is architectural — an evaluation step belongs before the revision step, not after. We release STET so that revision robustness, a model’s willingness to leave sufficient work alone, can be measured alongside first-turn capability. Intrinsic self-correction, on its own, is not enough: it requires extrinsic direction.

6 Limitations

Several limitations qualify these findings.

1. **Panel composition.** The balanced panel is dominated by Llama 3.3 70B (45 of 50 trials); other models exit the revision pool before Turn 5, so the only well-powered within-model estimate of the cliff is Llama’s. We do not claim all models degrade, only that among those that keep revising the powered comparison is significant.
2. **A single judge.** Quality is scored by one LLM judge (Claude Sonnet 4), chosen for the highest human correlation among six candidates but a single instrument nonetheless; human agreement is moderate ($\alpha = 0.529$), so ordinal means should defer to the better-validated binary sufficiency threshold.
3. **One temperature.** All generation uses temperature 1.0; revision behaviour may differ at lower temperatures.
4. **Task sample.** The forty tasks are a convenience sample across five domains, and per-domain Turn-5 cells are small ($n = 13$ to 30), so per-domain magnitudes are directional.
5. **Reversibility near chance.** After stripping, blind human preference for the first draft is 56%, a confidence interval that includes 50% and a pre-registered 65% bar that was not cleared; the pointwise degradation is real but not large enough to flip a blind preference once boilerplate is removed.
6. **An exploratory mechanism.** The input-dependent analysis (Figure 5) was not pre-registered; we report it as a mechanism, not a confirmatory result.
7. **No no-probe control.** The design cannot fully separate the revision act from the prompt that offers it; the bal-

anced probe, though built to be neutral, may still prime revision, making a no-probe control the stronger baseline for future work.

Ethics Statement

This study involved human evaluation by three raters (the first author and two research assistants) who scored AI-generated outputs on a quality scale. The evaluation involved reading and rating synthetic text only; no personal data was collected, and raters participated voluntarily. All data was generated by commercial LLM APIs in response to synthetic scenarios spanning code, data logic, analysis, and writing tasks, and the scenarios contain no sensitive content. Pricing data referenced in any cost estimate is drawn from publicly available API documentation.

References

- Anthropic. 2025a. The anthropic economic index.
- Anthropic. 2025b. Prompting best practices.
- Gagan Bansal, Tongshuang Wu, Joyce Zhou, Raymond Fok, Bismira Nushi, Ece Kamar, Marco Tulio Ribeiro, and Daniel Weld. 2021. Does the whole exceed its parts? the effect of AI explanations on complementary team performance. In *CHI*.
- Vadim Borisov, Michael Gröger, Mina Mikhael, and Richard H. Schreiber. 2026. Do chatbot LLMs talk too much? the Yap-Bench benchmark. *arXiv:2601.00624*.
- Zana Bućinca, Maja Barbara Malaya, and Krzysztof Z. Gajos. 2021. To trust or to think: cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making. *Proc. ACM HCI (CSCW)*.
- Guiming Hardy Chen, Shunian Chen, Ziche Liu, Feng Jiang, and Benyou Wang. 2024. Humans or LLMs as the judge? a study on judgement bias. In *EMNLP*. *arXiv:2402.10669*.
- Aaron Fanous, Jacob Goldberg, Ank A. Agarwal, Joanna Lin, Anson Zhou, Sonnet Xu, Vasiliki Biki, Roxana Daneshjou, and Sanmi Koyejo. 2025. SycEval: evaluating LLM sycophancy. In *AIES*, 8:893–900.
- Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. 2024. Large language models cannot self-correct reasoning yet. In *ICLR*. *arXiv:2310.01798*.
- Ryo Kamoi, Yusen Zhang, Nan Zhang, Jiawei Han, and Rui Zhang. 2024. When can LLMs actually correct their own mistakes? a critical survey of self-correction of LLMs. *TACL*, 12:1417–1440.
- Ryan Koo, Minhwa Lee, Vipul Raheja, Jong Inn Park, Zae Myung Kim, and Dongyeop Kang. 2024. Benchmarking cognitive biases in large language models as evaluators. In *Findings of ACL*, 517–545.
- Philippe Laban, Hiroaki Hayashi, Yingbo Zhou, and Jennifer Neville. 2026. LLMs get lost in multi-turn conversation. In *ICLR*. Outstanding Paper Award. *arXiv:2505.06120*.
- Dawei Li, Bohan Jiang, Liangjie Huang, Alimohammad Beigi, Chengshuai Zhao, Zhen Tan, Amrita Bhattacharjee, Yuxuan Jiang, Canyu Chen, Tianhao Wu, Kai Shu, Lu Cheng, and Huan Liu. 2025. From generation to judgment: opportunities and challenges of LLM-as-a-judge. In *EMNLP*, 2757–2791.

- Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, and 1 others. 2023. Holistic evaluation of language models. *TMLR*.
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, and 1 others. 2023. Self-refine: iterative refinement with self-feedback. In *NeurIPS*, 36:46534–46594.
- Moran Mizrahi, Guy Kaplan, Dan Malkin, Rotem Dror, Dafna Shahaf, and Gabriel Stanovsky. 2024. State of what art? a call for multi-prompt LLM evaluation. *TACL*, 12:933–949.
- OpenAI. 2025a. GPT-5 prompting guide.
- OpenAI. 2025b. Sycophancy in GPT-4o: what happened and what we’re doing about it.
- Arjun Panickssery, Samuel R. Bowman, and Shi Feng. 2024. LLM evaluators recognize and favor their own generations. In *NeurIPS*, 37:68772–68802.
- Raja Parasuraman and Dietrich H. Manzey. 2010. Complacency and bias in human use of automation: an attentional integration. *Human Factors*, 52(3):381–410.
- Ethan Perez, Sam Ringer, Kamile Lukošiušė, Karina Nguyen, Edwin Chen, Scott Heiner, Craig Pettit, Catherine Olsson, Sandipan Kundu, Saurav Kadavath, and 1 others. 2023. Discovering language model behaviors with model-written evaluations. In *Findings of ACL*, 13387–13434.
- Yuxiao Qu, Tianjun Zhang, Naman Garg, and Aviral Kumar. 2024. Recursive introspection: teaching language model agents how to self-improve. In *NeurIPS*. *arXiv:2407.18219*.
- Max Schemmer, Niklas Kuehl, Carina Benz, Andrea Bartos, and Gerhard Satzger. 2023. Appropriate reliance on AI advice: conceptualization and the effect of explanations. In *IUI*. *arXiv:2302.02187*.
- Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2024. Quantifying language models’ sensitivity to spurious features in prompt design. In *ICLR*. *arXiv:2310.11324*.
- Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R. Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R. Johnston, and 8 others. 2024. Towards understanding sycophancy in language models. In *ICLR*. *arXiv:2310.13548*.
- Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: language agents with verbal reinforcement learning. In *NeurIPS*, 36.
- Prasann Singhal, Tanya Goyal, Jiacheng Xu, and Greg Durrett. 2024. A long way to go: investigating length correlations in RLHF. In *COLM*. *arXiv:2310.03716*.
- Linda J. Skitka, Kathleen L. Mosier, and Mark Burdick. 1999. Does automation bias decision-making? *Int. J. Human-Computer Studies*, 51(5):991–1006.
- Romal Thoppilan, Daniel De Freitas, Jamie Hall, Noam Shazeer, Apoorv Kulshreshtha, Heng-Tze Cheng, Alicia Jin, Taylor Bos, Leslie Baker, Yu Du, and 1 others. 2022. LaMDA: language models for dialog applications. *arXiv:2201.08239*.
- Gladys Tyen, Hassan Mansoor, Victor Carbune, Peter Chen, and Tony Mak. 2024. LLMs cannot find reasoning errors, but can correct them given the error location. In *Findings of ACL*. *arXiv:2311.08516*.
- Helena Vasconcelos, Matthew Jörke, Madeleine Grundle-McLaughlin, Tobias Gerstenberg, Michael S. Bernstein, and Ranjay Krishna. 2023. Explanations can reduce overreliance on AI systems during decision-making. *Proc. ACM HCI (CSCW)*. *arXiv:2212.06823*.
- Jerry Wei, Da Huang, Yifeng Lu, Denny Zhou, and Quoc V. Le. 2023. Simple synthetic data reduces sycophancy in large language models. *arXiv:2308.03958*.
- Minghao Wu and Alham Fikri Aji. 2025. Style over substance: evaluation biases for large language models. In *COLING*. *arXiv:2307.03025*.
- Rongwu Xu, Brian S. Lin, Shujian Yang, Tianqi Zhang, Weiyan Shi, Tianwei Zhang, Zhixuan Fang, Wei Xu, and Han Qiu. 2024. The earth is flat because...: investigating LLMs’ belief towards misinformation via persuasive conversation. In *ACL*. *arXiv:2312.09085*.
- Chenyang Yang, Yike Shi, Qianou Ma, Michael Xieyang Liu, Christian Kästner, and Tongshuang Wu. 2026. What prompts don’t say: understanding and managing underspecification in LLM prompts. In *Findings of ACL*, 9072–9101.
- Jiayi Ye, Yanbo Wang, Yue Huang, Dongping Chen, Qihui Zhang, Nuno Moniz, Tian Gao, Werner Geyer, Chao Huang, Pin-Yu Chen, Nitesh V. Chawla, and Xiangliang Zhang. 2025. Justice or prejudice? quantifying biases in LLM-as-a-judge. In *ICLR*. *arXiv:2410.02736*.
- Xuanchang Zhang, Wei Xiong, Lichang Chen, Tianyi Zhou, Heng Huang, and Tong Zhang. 2025. From lists to emojis: how format bias affects model alignment. In *ACL*, 26940–26961. *arXiv:2409.11704*.
- Yizhe Zhang, Siqi Sun, Michel Galley, Yen-Chun Chen, Chris Brockett, Xiang Gao, Jianfeng Gao, Jingjing Liu, and Bill Dolan. 2020. DialoGPT: large-scale generative pre-training for conversational response generation. In *ACL: System Demonstrations*, 270–278.
- Yunxiang Zhang, Muhammad Khalifa, Lajanugen Logeswaran, Jaekyeom Kim, Moontae Lee, Honglak Lee, and Lu Wang. 2024. Small language models need strong verifiers to self-correct reasoning. In *Findings of ACL*. *arXiv:2404.17140*.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *NeurIPS*, 36:46595–46623.
- Enyu Zhou, Zhiheng Xi, Long Ma, Zhihao Zhang, Shihan Dou, Zhikai Lei, Guoteng Wang, Rui Zheng, Hang Yan, Tao Gui, Qi Zhang, and Xuanjing Huang. 2026. Steering LLMs via scalable interactive oversight. *arXiv:2602.04210*.

A Study 1: The Revision Gate

Study 1 tests whether user-stated quality thresholds constrain the decision to revise. In a fully factorial experiment (8 scenarios $\times$ 16 threshold conditions $\times$ 2 probes $\times$ 3 models $\times$ 5 runs = 3,840 trials), the follow-up probe’s phrasing dominates the revision gate. Under the leading probe (“Can this be improved?”), models revised 99.9% of the time regardless of threshold. Under the evaluative probe (“Take another look and let me know if it’s ready”), revision rates varied by model: Gemini 2.5 Flash declined 99.7%, GPT-4o 68.6%, Claude Sonnet 4 62.0%.

Chi-squared tests confirm the revision gate distribution differs significantly by probe type ($\chi^2 > 788$, $p < 0.0001$ for all models). Quality thresholds do not significantly affect the gate (Kruskal-Wallis $p > 0.40$ for all models). Within the leading-probe condition, higher thresholds produce moderately less overcorrection (Spearman ρ : Gemini -0.50 , Claude -0.35 , GPT-4o -0.14), but this calibration is locked behind the phrasing-sensitive gate.

Predictor	Model A	Model B	Model C
is_leading	3.47***	3.94***	—
threshold	−0.93***	−1.04***	−1.65***
is_qualitative	−0.74***	−0.85***	−1.05***
Claude Sonnet	0.10	0.15	0.33**
Gemini Flash	0.42***	0.52***	1.15***
Scenario FEs	No	Yes	Yes
<i>N</i>	3,840	3,840	1,920
AIC	5,319	4,839	3,545

Table 5: Ordinal regression (overcorrection $\sim$ predictors). Model C: leading-probe only. ** $p < 0.01$, *** $p < 0.001$. Reference: GPT-4o.

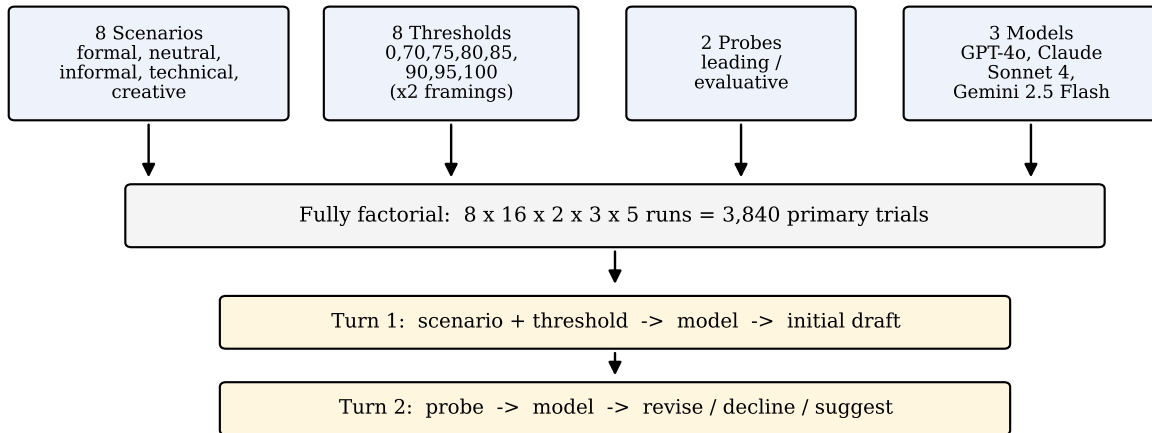


Figure 7: Study 1 experimental pipeline. Four factors crossed in a fully factorial design produce 3,840 trials.

Probe wordings and the shape of the revision gate. Across five probe wordings tested in Study 1 (3 models; $n = 3,932$), the revision gate is bimodal. The two probes that request change (“Can this be improved?” and “Is there anything you would change?”) trigger near-universal revision (99.9%, $n = 1,920$; 100%, $n = 18$). The three probes that request assessment produce single-digit to low-double-digit revision rates (23.2%, $n = 1,920$; 12.5%, $n = 24$; 2.0%, $n = 50$). No probe falls in the 40–99% range; the gate is semantic, not graded. The balanced probe, measured with a validated GENUINE/META classifier, lands in the assessment range at 39.3% at Turn 2 ($n = 720$).

B Study 2: Momentum

Study 2 tests whether prior revision rounds shift the revision gate on a subsequent evaluative probe (1,728 trials). Without prior revision (dose 0), models revised 23.2%; after 1–3 prior rounds, this rises to 44.6% ($\chi^2 = 376.54$, $p < 0.0001$). Model-level divergence is dramatic: GPT-4o jumps 31.4% $\rightarrow$ 98.4% at dose 1 (near-total sycophantic shift); Claude Sonnet 4 declines 38.0% $\rightarrow$ 24.5% at dose 3; Gemini 2.5 Flash shifts modestly 0.3% $\rightarrow$ 12.8% at dose 1. The momentum effect is a step function: the entire shift occurs between dose 0 and dose 1, with no additional gain at higher doses. Threshold level does not interact with momentum ($p = 0.51$).

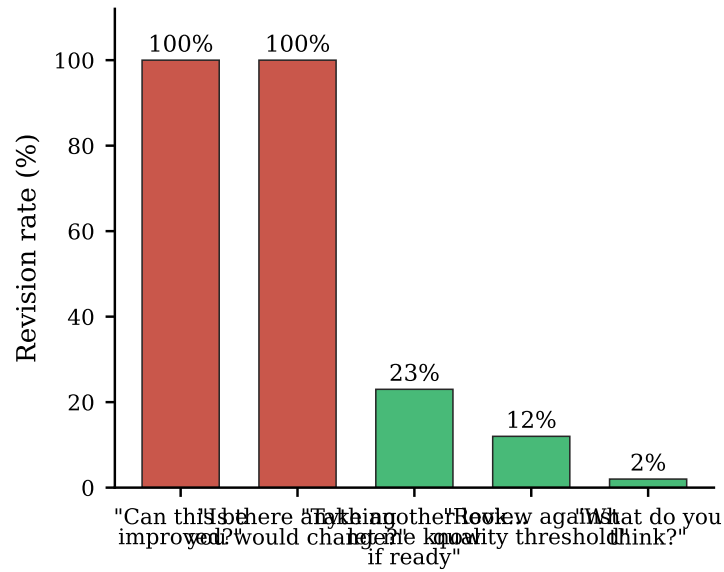


Figure 8: Compliance cliff (Study 1 pilot). Revision rates drop from 100% to $\leq 25\%$ across five pilot probes with no intermediate values.

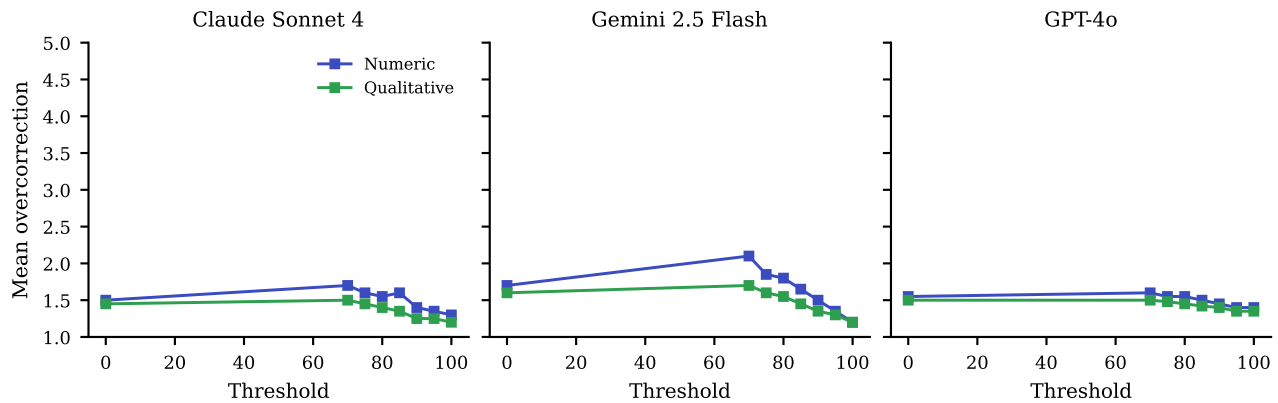


Figure 9: Overcorrection by threshold level within leading-probe trials (Study 1).

Reverse momentum. A single turn of affirming feedback (“This looks great, no changes needed”) suppresses full revision to near-zero across all models: Gemini 0.0%, GPT-4o 1.0%, Claude 21.9% (all minor suggestions). The gate tracks the most recent conversational signal in both directions.

C Pipeline Diagrams

See Figure 7 and Figure 2.

D Supplementary Figures

See Figures 8, 9, and 10.

E Full Statistical Tables

E.1 Pairwise Threshold Comparisons (Study 1)

E.2 Power Analysis

Post-hoc power analysis at $\alpha = 0.05$, 80% power. Study 1 RQ1 ($N = 1,920$ per probe): MDE for proportion difference = 0.032; observed ≈ 0.75 , massively overpowered. Study 1 RQ2 ($N = 1,920$ leading-probe): MDE for $|\rho| = 0.064$; all observed correlations exceed this. Study 3 ($N = 720$, balanced panel $n = 50$): panel-level MDE = 0.5 levels; observed $\Delta = 0.94$ unstripped (0.74 stripped), both exceeding the MDE. The balanced panel yields $p = 3.32 \times 10^{-6}$ (Wilcoxon, $r = 0.658$) on the unstripped cliff; the stripped cliff ($\Delta = 0.74$) has $p = 1.01 \times 10^{-4}$ ($r = 0.55$).

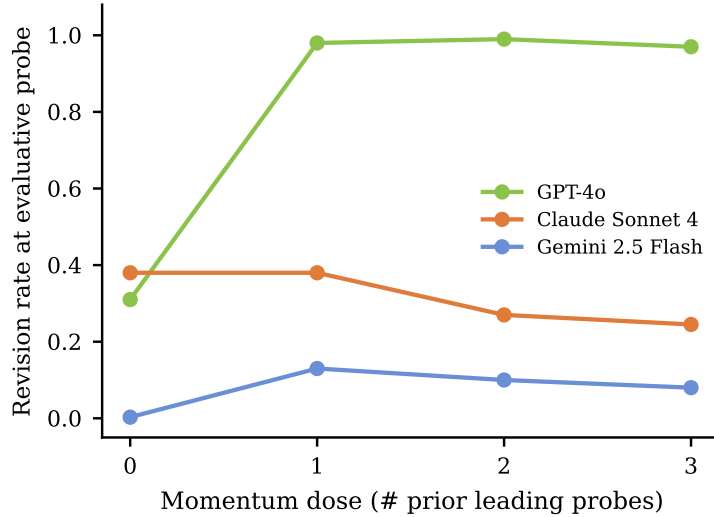


Figure 10: Revision rate by momentum dose (Study 2). The momentum effect is a step function driven by GPT-4o.

Model	Comparison	U	p_{adj}	r
Gemini (num.)	70 vs. 100	4935	<0.001	-0.42
Gemini (num.)	85 vs. 100	4238	<0.001	-0.32
Gemini (num.)	0 vs. 100	4755	0.002	-0.25
Gemini (qual.)	0 vs. 100	4084	0.002	-0.28
Claude (num.)	85 vs. 100	4146	<0.001	-0.30
Claude (qual.)	85 vs. 100	3880	0.024	-0.21

Table 6: Significant pairwise comparisons (Bonferroni-corrected, $q < 0.05$).

F Judge Calibration Details

Candidate judge	Spearman r	QW κ
Claude Sonnet 4	0.505	0.526
DeepSeek V4	0.340	0.398
Qwen 3 235B	0.272	0.017
Llama 3.3 70B	0.181	0.022
GPT-4o	0.140	0.065
Gemini 2.5 Flash	-0.036	-0.026

Table 7: Judge calibration ($n = 64$ stratified samples; Spearman correlation against the human rater mean). Claude Sonnet 4 was selected ($r = 0.505$, $p = 2.1 \times 10^{-5}$).

F.1 Human Rater Agreement

G Study 1 Inter-Rater Reliability

H The Trial Shown in Figure 1

The complete exchange for the quarterly sales trial (Llama 3.3 70B, run 3). Text is shown after meta-commentary stripping, the basis on which every quality score is computed; raw responses are longer at turns two through five. Judge rationales are quoted verbatim.

Task (Turn 1). *I need to write up our quarterly sales for a management meeting. Q1 revenue was \$2.1M, Q2 \$2.4M, Q3 \$1.9M, Q4 \$3.1M. Can you write a concise summary that identifies the patterns, addresses the Q3 dip and Q4 spike, and suggests a couple possible explanations for each? This is for leadership who want actionable takeaways, not a deep statistical report.*

Probe (repeated verbatim, Turns 2–5). *Would you like to keep this as your final version, or would you like to revise it?*

Turn 1 (Level 4, 214 words). A full summary reporting each quarter, the 21% Q3 dip with two candidate explanations (seasonal slowdown; new-competitor share loss) and the 63% Q4 spike with three (holiday promotions; year-end strategies; product launch),

Rater pair	QW κ	Binary	Within 1
Liam – Sophie	0.578	78.1%	95.3%
Sophie – Troy	0.603	82.8%	87.5%
Liam – Troy	0.406	76.6%	81.2%
All three (α)	0.529	—	—
Evaluator – Human	0.526	—	—

Table 8: Human rater agreement on 64 stratified samples (6→2 recode applied). Evaluator–Human mean: Spearman $r = 0.505$ ($p < 0.001$).

Dimension	QW κ	Interpretation
Revision magnitude	0.844	Excellent
Revision value	0.690	Good
Overcorrection	0.609	Acceptable
Threshold alignment	0.556	Marginal

Table 9: Inter-rater reliability for Study 1 (GPT-4o vs. Claude Sonnet 4, $n = 60$).

closing with recommendations. *Judge:* “addresses all requested components with appropriate tone for leadership, though the explanations are somewhat generic.”

Turn 2 (Level 4, 125 words). A shorter version retaining both explanations but adding repetitive recommendation language. *Judge:* “exceeds the request for a concise summary by adding lengthy recommendations and repetitive language that drifts from the original ask.”

Turn 3 (Level 3, 95 words). Further compressed; the specific explanations begin to thin. *Judge:* “lacks specificity in the explanations and doesn’t clearly present the actual quarterly numbers.”

Turn 4 (Level 2, 79 words). The explicitly requested explanations for the Q3 dip and Q4 spike are gone, replaced by generic recommendations. *Judge:* “missing the explicitly requested explanations for the Q3 dip and Q4 spike, providing only generic recommendations.”

Turn 5 (Level 2, 79 words). Essentially identical to Turn 4; the requested explanations do not return. *Judge:* “missing the explicitly requested specific explanations, providing only generic recommendations.”

I Secondary Result Tables

Dimension	QW κ	% Agree	% Within 1
Revision magnitude	0.575	68.3	85.0
Revision value	0.669	76.7	88.3
Overcorrection	0.392	71.7	90.0
Threshold alignment	0.339	38.3	85.0

Table 10: Inter-rater reliability for Study 2 ($n = 60$).

	n	% down	p
<i>By model</i>			
Llama 3.3 70B	353	68.9	<0.001
Claude Sonnet 4	196	58.5	0.11
Qwen 3 235B	90	74.5	<0.001
GPT-4o	40	80.0	0.006
DeepSeek V4	31	85.7	0.007
Gemini 2.5 Flash	8	100.0	0.062
<i>By domain</i>			
Writing	118	79.1	<0.001
Creative	145	73.1	<0.001
Code	196	67.0	<0.001
Analysis	125	65.9	0.024
Data logic	134	65.9	0.024

Table 11: Direction of genuine revisions on stripped content. n counts genuine revisions; the percentage is over the subset whose level moves. Sign test, one-sided. A χ^2 test of homogeneity across the five domains does not reject uniformity ($\chi^2 = 3.02$, $df = 4$, $p = 0.555$).

Turn	Stripped (Unstripped)	Shift	n
T1	4.11 (4.11)	0	720
T2	3.66 (3.25)	+0.41	283
T3	3.40 (3.01)	+0.39	185
T4	3.34 (2.97)	+0.37	154
T5	3.07 (2.85)	+0.22	96
Δ (T1–T5)	−1.04 (−1.26)		

Table 12: Pooled mean quality by turn, restricted to genuine revisions (all 720 at T1). Stripped scores primary. n declines as models exit the revision pool.

Domain	Stripped	Unstr.	n (T5)
analysis	−1.16	(−1.21)	17
code	−1.05	(−1.15)	30
creative	−0.82	(−1.22)	19
data_logic	−1.01	(−1.35)	17
writing	−1.06	(−1.31)	13

Table 13: Quality decline (T1–T5) by domain on stripped content. All domains show negative deltas; 4/5 significant ($p < 0.05$), data_logic marginal ($p = 0.058$). Per-domain n (T5) limits precision.